Egypt Residential Real Estate Power BI

Project Planning & Management

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa

1. Project Proposal

Overview. Build an interactive Power BI report on Egyptian residential real estate transactions, using a star schema: 1 fact table (about 25,000 transactions) and 8 dimensions (Date, Property, PropertyType, Location, Developer, Agent, Customer, Channel).

Objectives

- Give stakeholders a clear view of sales, pricing, cancellations, and agent/channel performance.
- Deliver a validated, well-modeled dataset with documented metric definitions.
- Publish a polished, navigable report to the agreed workspace.

Scope

- *In scope:* data cleaning, star-schema modeling, DAX measures, report pages, interactivity, validation, publishing, documentation.
- *Out of scope:* live data feeds, predictive models, external data sources.

2. Project Plan

Timeline (Gantt, 6 weeks - adjust to the actual deadline)

Phase	Owner	W1	W2	W3	W4	W5	W6
Requirements & kickoff	Sama Raed						
Data preparation (Power Query)	Saleh						
Data modeling	Sama Raed						
DAX measures & metrics	Maged						
Report design & KPI cards	Abd Elkhalek						
Interactivity & navigation	Mostafa						
Quality review	Maged						
Validation & publishing	Sama Abdelmeged						
Documentation & handover	Sama Abdelmeged						
Final presentation	Abd Elkhalek						

Milestones

- M1 (end W1): requirements approved, data loaded.
- M2 (end W2): clean data ready.
- M3 (end W3): relationships validated.
- M4 (end W4): core measures complete.
- M5 (end W5): full report draft with interactivity.
- M6 (end W6): validated, published, presented.

Deliverables: requirements document, cleaned dataset, data model, DAX measure library, Power BI report (.pbix), validation checklist, metric definitions, handover pack, final presentation.

Resources: Power BI Desktop and Service, Power Query, DAX, the 10 source CSV files, shared workspace/drive, weekly team meeting.

3. Task Assignment & Roles

Member	Technical responsibility	Non-technical role	Key outputs
Saleh	Power Query & data preparation: import, clean, transform; check data types and missing values	Project Coordinator: track tasks, deadlines, progress	Clean tables, data-quality log, task tracker
Maged	DAX & metrics: measures and calculated columns for key metrics and trends	Quality Reviewer: check clarity, consistency, ease of understanding	Measure library, review feedback list
Abd Elkhalek	Report design: pages, visuals, KPI cards	Team Leader & Presenter: coordinate team, present findings	Report pages, final presentation
Sama Raed	Data modeling: create and validate fact-dimension relationships	Requirements Lead: gather and organize stakeholder needs	Requirements doc, star-schema model
Sama Abdelmeged	Report validation & publishing: verify measures/visuals, publish to workspace	Documentation Lead: project notes, metric definitions, handover	Validation checklist, published report, handover pack
Mostafa	Interactivity & navigation: slicers, filters, drill-through, navigation	Meeting Facilitator: organize discussions, record action items	Interactive report, meeting minutes

Handoff flow: Sama Raed (requirements) -> Saleh (data) -> Sama Raed (model) -> Maged (DAX) -> Abd Elkhalek and Mostafa (design and interactivity) -> Maged (review) -> Sama Abdelmeged (validate and publish) -> Abd Elkhalek (present).

4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation	Owner
1	Data quality issues (nulls, wrong types, duplicates)	Medium	High	Profile data early; keep a data-quality log; use the validation report	Saleh
2	Wrong or broken relationships (duplicated keys, ambiguous filters)	Medium	High	Validate cardinality and filter direction; test with sample totals	Sama Raed
3	Incorrect DAX (e.g., mixing cancelled and successful deals)	Medium	High	Document each definition; cross-check vs source totals; peer review	Maged
4	Slow report performance with 25K rows and many visuals	Low	Medium	Limit visuals per page; prefer measures over calculated columns; remove unused columns	Abd Elkhalek
5	Cluttered or inconsistent design	Medium	Medium	Agree on a theme and layout template; quality review before validation	Maged / Abd Elkhalek
6	Broken slicers, drill-through, or navigation	Medium	Medium	Test every interaction on each page; keep a test checklist	Mostafa
7	Requirements change or are unclear	Medium	Medium	Sign off requirements in W1; log changes with approval	Sama Raed
8	Missed deadlines or unclear task ownership	Medium	High	Weekly tracker review; early escalation of blockers	Saleh
9	Publishing or workspace access problems	Low	High	Confirm workspace access in W2; trial publish in W4	Sama Abdelmeged
10	Missing or outdated documentation	Medium	Medium	Update docs weekly; finalize metric definitions at M4	Sama Abdelmeged
11	Poor communication and unrecorded decisions	Low	Medium	Weekly meeting with minutes and action items	Mostafa

5. KPIs

A. Project success KPIs

KPI	Target	Owner
Milestones delivered on time	>= 90%	Saleh
Tasks completed vs planned (weekly)	>= 90%	Saleh
Data quality: null/invalid values after cleaning	0 in key columns	Saleh
Relationship validation: tables correctly related	100%	Sama Raed
Measure accuracy vs source totals	100% match	Sama Abdelmeged
Report page load / visual response time	< 5 seconds	Abd Elkhalek
Interaction test pass rate (slicers, drill-through, navigation)	100%	Mostafa
Quality review issues closed before publishing	100%	Maged
Meeting action items closed on time	>= 85%	Mostafa
Report availability in workspace (uptime)	>= 99%	Sama Abdelmeged
Stakeholder user adoption (viewers within first month)	>= 80% of target users	Abd Elkhalek
Stakeholder satisfaction (clarity and usefulness)	>= 4 / 5	Sama Raed

B. Report business KPIs (built by Maged, displayed by Abd Elkhalek)

- Total transactions and total net sales (EGP)
- Average price per sqm (EGP)
- Successful sale rate and cancellation rate
- Average days to close
- Average discount rate
- Total commission (EGP)
- Sales by location, property type, developer, agent, and channel
- Monthly and yearly sales trend

6. Summary

Six members deliver a Power BI real estate report in about 6 weeks. Saleh prepares data and coordinates, Sama Raed defines requirements and the model, Maged builds the DAX and reviews quality, Abd Elkhalek designs and presents, Mostafa adds interactivity and runs meetings, and Sama Abdelmeged validates, publishes, and documents. Top risks are data quality, model and DAX errors, and deadlines, each with a named owner. Success is tracked through on-time delivery, 100% measure accuracy, fast response time, high uptime, and strong user adoption.

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Literature Review: Feedback, Improvements & Grading

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa

1. Feedback & Evaluation

Purpose. Record the lecturer's assessment of the project and map each comment to the team member responsible. Fields marked *Lecturer* are left open until feedback is received. The *Team self-check* column is the team's own pre-review assessment.

Evidence from the project (validation report): all 11 data checks passed, including 25,000 unique transactions, full key coverage for property, location, and date, correct amount and balance logic, zero net amount for cancelled deals, and valid Egyptian geography.

Area	Team self-check	Lecturer's assessment	Responsible
Data preparation	Strong: all validation checks passed	*Lecturer*	Saleh
Data model	Star schema with 1 fact and 8 dimensions; relationships validated	*Lecturer*	Sama Raed
DAX & metrics	Core KPIs defined; definitions documented	*Lecturer*	Maged
Report design	Clear KPI cards and consistent theme	*Lecturer*	Abd Elkhalek
Interactivity	Slicers, drill-through, and navigation tested	*Lecturer*	Mostafa
Validation & publishing	Totals checked against source; report published	*Lecturer*	Sama Abdelmeged
Requirements & scope	Stakeholder needs documented	*Lecturer*	Sama Raed
Project management	Milestones and tasks tracked	*Lecturer*	Saleh
Documentation	Metric definitions and handover pack	*Lecturer*	Sama Abdelmeged
Presentation	Findings presented clearly	*Lecturer*	Abd Elkhalek

Overall comment (Lecturer): _____

2. Suggested Improvements

#	Improvement	Benefit	Owner
1	Add automated data-quality checks in Power Query (null, duplicate, and range rules that run on every refresh)	Problems are caught at refresh, not by manual review	Saleh
2	Add a formal data dictionary covering every table and column	Easier onboarding and handover	Sama Abdelmeged
3	Add time intelligence measures (year-over-year, month-over-month, running totals)	Stronger trend analysis	Maged
4	Add a measure-testing table that compares each DAX result with a source total	Proves measure accuracy	Maged
5	Add an executive summary page and a drill-down page for locations and developers	Faster insight for managers	Abd Elkhalek
6	Add bookmarks, tooltips, and a reset-filters button	Better usability	Mostafa
7	Add accessibility checks (contrast, alt text, tab order) and mobile layout	Wider audience	Abd Elkhalek
8	Add row-level security by region or agent office if required	Controlled access to data	Sama Abdelmeged
9	Set up scheduled refresh and report usage monitoring	Supports uptime and adoption KPIs	Sama Abdelmeged
10	Run a short user feedback session and log changes	Evidence-based improvement	Sama Raed
11	Keep meeting minutes and a change log in one shared place	Traceable decisions	Mostafa
12	Run a peer review checklist before each release	Consistent quality	Maged

Priority: high = 3, 4, 9; medium = 1, 2, 5, 8; low = 6, 7, 10, 11, 12.

3. Final Grading Criteria

Proposed breakdown. Replace with the official course rubric if it differs.

Category	Marks	What is assessed	Main contributors
Documentation	25	Proposal, plan, roles, risks, KPIs, metric definitions, handover	Sama Abdelmeged, Sama Raed, Saleh
Implementation	40	Data preparation, star-schema model, DAX measures, report design, interactivity	Saleh, Sama Raed, Maged, Abd Elkhalek, Mostafa
Testing	15	Data validation, measure accuracy, interaction tests, review and publishing checks	Sama Abdelmeged, Maged, Mostafa
Presentation	20	Clarity, structure, demo of the report, answers to questions	Abd Elkhalek (lead), whole team
Total	**100**		

Detailed marking guide

Category	Criterion	Marks
Documentation	Completeness and structure	10
Documentation	Metric definitions and data dictionary	8
Documentation	Clarity of roles, risks, and KPIs	7
Implementation	Data cleaning and preparation	8
Implementation	Data model quality	8
Implementation	DAX measures and accuracy	10
Implementation	Report design and KPI cards	8
Implementation	Interactivity and navigation	6
Testing	Data and relationship validation	5
Testing	Measure and visual verification	5
Testing	Interaction tests and publishing check	5
Presentation	Content and storytelling	8
Presentation	Demo and delivery	7
Presentation	Q&A and teamwork	5

Mark record

Category	Marks available	Marks awarded	Lecturer comment
Documentation	25		
Implementation	40		
Testing	15		
Presentation	20		
Total	**100**		

4. Summary

The project's data checks all pass and each member owns a clear area. The lecturer's assessment is left open for entry. The 12 suggested improvements focus on measure testing, time intelligence, scheduled refresh, and documentation, and each has an owner. Marks are proposed as Documentation 25, Implementation 40, Testing 15, and Presentation 20, and each category is linked to the members who deliver it.

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Requirements Gathering

Team: Saleh, Maged, Abd Elkhalek, Sama Raed, Sama Abdelmeged, Mostafa. Requirements Lead: Sama Raed.

1. Stakeholder Analysis

Stakeholder	Interest	Key needs	Influence	Team contact
Sales / executive managers	Overall sales performance	Total net sales, trends, cancellation rate, top locations and developers	High	Abd Elkhalek
Sales team leaders	Agent and office performance	Sales by agent, agent level, office region, and channel; commission totals	High	Maged
Agents	Own results and targets	Own sales, discounts, days to close	Medium	Mostafa
Marketing / channel managers	Channel effectiveness	Sales and commission rate by channel; customer segment and nationality mix	Medium	Sama Raed
Pricing / market analysts	Price levels and demand	Price per sqm by location, market tier, property type, finishing level	Medium	Maged
Developer relations	Developer performance	Sales and cancellations by developer and developer tier	Medium	Sama Raed
Finance team	Revenue and receivables	Gross vs net amount, down payment, balance due, payment method	Medium	Sama Abdelmeged
Lecturer / evaluators	Project quality	Complete documentation, correct metrics, clear presentation	High	Abd Elkhalek
Project team	Delivery	Clear scope, clean data, defined metrics	High	Saleh

Gathering methods: stakeholder interviews and short surveys (Sama Raed), review of the 10 source files and validation report (Saleh), requirements workshop with sign-off in week 1 (Sama Raed, notes by Mostafa), and a feedback session before publishing.

2. User Stories & Use Cases

User stories

ID	As a...	I want...	So that...	Priority
US-01	Sales manager	a summary page with total net sales, transactions, and cancellation rate	I see overall performance at once	High
US-02	Sales manager	to compare sales by governorate, city, and district	I find the strongest and weakest markets	High
US-03	Team leader	to rank agents by sales and commission	I recognize top performers	High
US-04	Agent	to see my own sales, discounts, and days to close	I track my targets	Medium
US-05	Marketing manager	to compare channels by sales and commission rate	I invest in the best channels	Medium
US-06	Pricing analyst	to view average price per sqm by location, property type, and finishing level	I judge price levels	High
US-07	Developer relations	to see sales and cancellations by developer	I assess partners	Medium
US-08	Finance officer	to see gross amount, discount, net amount, down payment, and balance due	I follow revenue and receivables	Medium
US-09	Any user	to filter by year, month, location, property type, and status	I focus on what matters to me	High
US-10	Any user	to drill from a summary to transaction details	I understand the reason behind a number	Medium
US-11	Any user	to navigate pages with clear buttons and a visible reset-filters option	I use the report without training	Medium
US-12	Lecturer	to read clear metric definitions and the project documentation	I assess the work fairly	High

Use cases

ID	Use case	Actor	Main flow	Result
UC-01	View sales overview	Sales manager	Open report; summary page loads; select a year with the slicer; KPI cards update	Manager sees filtered KPIs
UC-02	Analyze a market	Sales manager	Choose a governorate; map and charts update; drill through to district and property type	Manager identifies top districts
UC-03	Review agent performance	Team leader	Open agent page; filter by office region and agent level; sort by sales	Ranked agent list with commission
UC-04	Compare channels	Marketing manager	Open channel page; compare sales and commission rate; filter by customer segment	Best channel identified
UC-05	Check finance position	Finance officer	Open finance page; filter by payment method; view net amount and balance due	Receivables visible
UC-06	Investigate cancellations	Any user	Select Cancelled status; drill through to transactions list	Cancellation causes explored
UC-07	Refresh and publish	Sama Abdelmeged	Refresh data; run validation checks; publish to workspace	Updated report available

3. Functional Requirements

ID	Requirement	Linked stories	Owner	Priority
FR-01	Import the fact table and all 8 dimension tables and clean data types and missing values	All	Saleh	High
FR-02	Model a star schema with one-to-many relationships from each dimension to the fact table	All	Sama Raed	High
FR-03	Provide measures for total transactions, gross amount, net amount, average price per sqm, average discount, down payment, balance due, and commission	US-01, US-03, US-08	Maged	High
FR-04	Provide successful sale rate, cancellation rate, and average days to close	US-01, US-10	Maged	High
FR-05	Provide time intelligence: year-over-year and month-over-month change, running total	US-01	Maged	Medium
FR-06	Create report pages: overview, locations, property types, developers, agents, channels, finance, details	US-01 to US-08	Abd Elkhalek	High
FR-07	Display KPI cards, trend lines, bar charts, map, and tables with a consistent theme	US-01, US-02	Abd Elkhalek	High
FR-08	Provide slicers for year, month, location, property type, developer, channel, and transaction status	US-09	Mostafa	High
FR-09	Provide drill-through from summary visuals to transaction details	US-10	Mostafa	Medium
FR-10	Provide navigation buttons, bookmarks, and a reset-filters button	US-11	Mostafa	Medium
FR-11	Verify every measure against source totals before release	All	Sama Abdelmeged	High
FR-12	Publish the report to the agreed workspace	UC-07	Sama Abdelmeged	High
FR-13	Provide metric definitions, data dictionary, and handover notes	US-12	Sama Abdelmeged	High
FR-14	Review the report for clarity and consistency before validation	US-11, US-12	Maged	Medium
FR-15	Record requirements, changes, and meeting action items	All	Sama Raed / Mostafa	Medium

4. Non-functional Requirements

ID	Category	Requirement	Measure	Owner
NFR-01	Performance	Report pages load and respond quickly with 25,000 transactions	Under 5 seconds per page and visual update	Abd Elkhalek
NFR-02	Performance	Efficient model: measures over calculated columns, unused columns removed	Model size and refresh time tracked	Saleh / Maged
NFR-03	Security	Only authorized users can open the report	Workspace access limited to approved users	Sama Abdelmeged
NFR-04	Security	Row-level security by region or office when required	Tested with sample roles	Sama Abdelmeged
NFR-05	Security	Customer data is limited to what the analysis needs	Sensitive fields reviewed before publishing	Saleh
NFR-06	Usability	Clear layout, readable fonts, consistent colors, and tooltips	Quality review checklist passed	Maged
NFR-07	Usability	New users can find key figures without training	At least 4 of 5 in a short user test	Sama Raed
NFR-08	Usability	Accessible design: good contrast, alt text, tab order	Accessibility check passed	Abd Elkhalek
NFR-09	Reliability	Report is available in the workspace	Availability of at least 99%	Sama Abdelmeged
NFR-10	Reliability	Data refresh completes without errors	Scheduled refresh succeeds	Saleh
NFR-11	Reliability	Measures return accurate results	100% match with source totals	Maged
NFR-12	Maintainability	Measures and tables are named and documented	Data dictionary complete	Sama Abdelmeged

5. Summary

Nine stakeholder groups need clear sales, price, agent, channel, developer, and finance views from 25,000 transactions. Twelve user stories and seven use cases describe how they work with the report. Fifteen functional requirements cover data preparation, modeling, measures, report pages, interactivity, validation, publishing, and documentation, each with a named owner. Twelve non-functional requirements set targets for performance (under 5 seconds), security, usability, and reliability (99% availability, 100% measure accuracy).